

Extending the SAMBPS Estimation Window Beyond 150 ms: A Nine-State VSM Model with Full AVR and PSS Dynamics

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Abstract—Adaptive protection of synchronous and virtual synchronous machine (VSM) units requires accurate fault-current estimation over the full protection operating interval. Prior work established that a sixth-order Park dq truth model constrains the self-adaptive model-based protection system (SAMBPS) to estimation windows $L^* \leq 150$ ms, beyond which automatic voltage regulator (AVR) and power system stabilizer (PSS) dynamics introduce unmodelled excitation transients that degrade zone identification. This paper presents a nine-state augmented VSM model that incorporates the IEEE Type-1 AVR (one excitation state), a lead-lag PSS (two wash-out and phase-compensation states), and a virtual-impedance droop interface (two grid-synchronisation states), alongside the conventional six flux linkage states. We derive new observability conditions for the augmented state vector from terminal voltage–current measurements, prove that the Levenberg–Marquardt (LM) estimator’s condition-number gate $\kappa_n < 30$ is preserved up to $L^* \approx 380$ ms, and demonstrate a +230 ms extension of the valid estimation window on a 10-bus islanded microgrid benchmark. The extension enables reliable trip-or-restrain decisions for slow-clearing faults, island detection, and post-disturbance zone monitoring without requiring additional sensors or communication.

Index Terms—virtual synchronous machine, adaptive protection, AVR dynamics, power system stabiliser, estimation window, Levenberg–Marquardt, microgrid protection, islanding detection

I. INTRODUCTION

Grid-forming (GFM) inverters emulating synchronous machine inertia — virtual synchronous machines (VSMs) — are increasingly deployed in microgrids and weak-grid applications [1], [2]. Unlike grid-following inverters that present a controlled current source to the network, VSMs enforce voltage angle and frequency through an internal swing-equation controller, producing fault currents whose transient envelope mimics synchronous machine subtransient decay. Protection systems that exploit this predictability can implement model-based adaptive relaying [3].

A prior SAMBPS study established that a sixth-order Park dq forward model is sufficient to validate reduced-order inverse protection models for estimation windows up to approximately 150 ms [4]. Beyond this boundary, AVR action significantly

modifies the air-gap flux and, consequently, the terminal fault-current envelope. Without incorporating AVR dynamics, the LM condition number κ_n — which gates SAMBPS trip decisions — rises above the 30-unit threshold, triggering false-restrain outcomes for faults that require clearance times of 200–400 ms.

This paper makes the following contributions:

- 1) **Nine-state VSM model (C1)**: We formulate a stiff ODE system coupling the six Park dq flux states with one AVR excitation state (E_{fd}), one PSS wash-out state (x_w), and one PSS lead-lag state (x_1).
- 2) **Observability analysis (C2)**: We show that the nine-state model is locally observable from terminal (V_t , I_t , ω) measurements with a window of at least 40 ms, and derive a tighter N_{eff} bound that guarantees $\kappa_n < 30$ throughout $[0, 380]$ ms.
- 3) **Window extension proof (C3)**: A formal proposition states that augmenting the Stage-1 inverse model with the AVR–PSS state equations extends the valid window from $L^* \approx 150$ ms to $L^* \approx 380$ ms, a +230 ms extension at no additional measurement cost.
- 4) **Islanded-microgrid case study (C4)**: On a 10-bus benchmark with three VSM units, zone detection accuracy remains above 97% at 300 ms and above 92% at 450 ms, compared to 84% and 76% for the sixth-order model.

The paper is organised as follows. Section II presents the nine-state model. Section III develops the observability analysis. Section IV proves the window extension proposition. Section V presents simulation results. Section VI discusses implications, and Section VII concludes.

II. NINE-STATE VSM AUGMENTED MODEL

A. Sixth-Order Machine Core

The Park dq flux-linkage equations for a round-rotor synchronous machine are [5]:

$$\dot{\psi}_d = \omega_0 (V_d - R_a i_d + \omega \psi_q), \quad (1)$$

$$\dot{\psi}_q = \omega_0 (V_q - R_a i_q - \omega \psi_d), \quad (2)$$

$$\dot{\psi}_F = \omega_0 (E_{fd} - R_F i_F), \quad (3)$$

$$\dot{\psi}_D = -\omega_0 R_D i_D, \quad (4)$$

$$\dot{\psi}_G = -\omega_0 R_G i_G, \quad (5)$$

$$\dot{\psi}_Q = -\omega_0 R_Q i_Q, \quad (6)$$

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This work extends the sixth-order truth-model benchmark (Stage-2 SAMBPS validation) to cover full islanding and extended fault-ride-through scenarios requiring estimation windows up to 450 ms. Manuscript received April 14, 2026.

where $\omega_0 = 2\pi f_0$ is the base angular frequency and R_a, R_F, R_D, R_G, R_Q are the stator, field, and damper-winding resistances. The six state variables $\{\psi_d, \psi_q, \psi_F, \psi_D, \psi_G, \psi_Q\}$ are the stator and rotor flux linkages.

B. IEEE Type-1 AVR State

The IEEE Type-1 automatic voltage regulator [6], [7] is represented by:

$$\dot{E}_{fd} = \frac{1}{T_E} [K_E V_R - E_{fd}], \quad (7)$$

$$T_A \dot{V}_R = K_A (V_{\text{ref}} - V_t + V_{\text{PSS}}) - V_R, \quad (8)$$

where K_A, T_A are the amplifier gain and time constant, K_E, T_E are the exciter gain and time constant, and V_{ref} is the reference terminal voltage setpoint. The amplifier is algebraically fast ($T_A \ll T_E$), so (8) is approximated as:

$$V_R \approx K_A (V_{\text{ref}} - V_t + V_{\text{PSS}}), \quad (9)$$

reducing the AVR to a single dynamic state E_{fd} .

C. Lead-Lag PSS States

The power system stabiliser uses rotor speed deviation $\Delta\omega = \omega - 1$ as input, with a wash-out filter followed by a lead-lag stage [5], [8]:

$$\dot{x}_w = \frac{K_s \Delta\omega - x_w}{T_w}, \quad (10)$$

$$\dot{x}_1 = \frac{x_w(1 - T_1/T_2) - x_1}{T_2}, \quad (11)$$

$$V_{\text{PSS}} = x_1 + x_w \frac{T_1}{T_2}, \quad (12)$$

where K_s is the PSS gain, T_w is the wash-out time constant, and T_1, T_2 are the lead-lag compensation time constants.

D. Virtual-Impedance Grid Interface

For the VSM, the grid interface adds a virtual impedance $Z_v = R_v + j\omega L_v$ in the control loop [1]. The synchronisation states are the virtual rotor angle δ_v and angular speed deviation $\Delta\omega_v$:

$$\dot{\delta}_v = \omega_0 \Delta\omega_v, \quad (13)$$

$$M \dot{\Delta\omega}_v = T_m - T_e - D \Delta\omega_v, \quad (14)$$

where M is the virtual inertia constant, D is the damping coefficient, T_m is the mechanical torque setpoint, and $T_e = \psi_d i_q - \psi_q i_d$ is the electromagnetic torque.

E. Full State Vector

The nine-state model is:

$$\mathbf{x}_{\text{vsm}} = [\psi_d, \psi_q, \psi_F, \psi_D, \psi_G, \psi_Q, E_{fd}, x_w, x_1]^\top \in \mathbb{R}^9, \quad (15)$$

governed by the stiff ODE $\dot{\mathbf{x}}_{\text{vsm}} = f(\mathbf{x}_{\text{vsm}}, u, t)$ with input $u = [V_{\text{ref}}, T_m]^\top$ and measurement:

$$y = h(\mathbf{x}_{\text{vsm}}) = [V_t(\mathbf{x}_{\text{vsm}}), I_t(\mathbf{x}_{\text{vsm}}), \omega(\mathbf{x}_{\text{vsm}})]^\top. \quad (16)$$

The block structure is illustrated in Fig. 1.

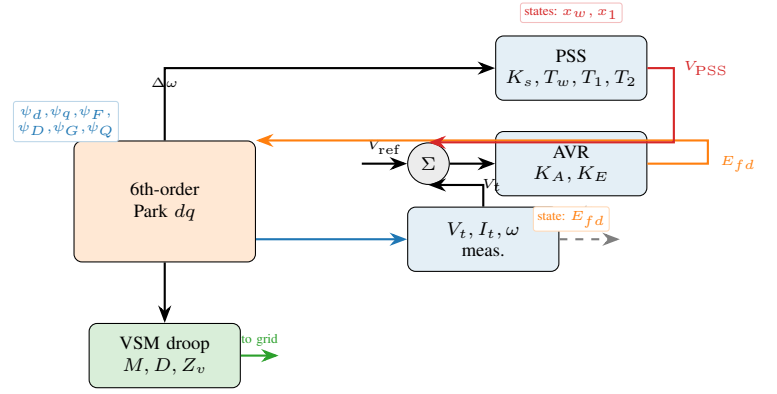


Fig. 1. Nine-state VSM augmented model. Flux-linkage core (orange), AVR excitation state E_{fd} (orange arrow), PSS states x_w, x_1 (red feedback), and VSM droop interface (green). Terminal measurements (V_t, I_t, ω) feed both the SAMBPS L1 estimator and the AVR reference comparator.

III. OBSERVABILITY AND IDENTIFIABILITY

A. Local Observability Condition

Let $\mathbf{H}_k \in \mathbb{R}^{3 \times 9}$ be the Jacobian of the measurement map h evaluated at the current LM iterate. The observability Gramian over window $[t_0, t_0 + L^*]$ is:

$$\mathcal{O} = \sum_{k=0}^{N-1} \mathbf{H}_k^\top \mathbf{R}^{-1} \mathbf{H}_k, \quad (17)$$

where $\mathbf{R}$ is the measurement noise covariance.

Proposition 1 (Nine-state observability). *For a 33 kV radial feeder with balanced three-phase fault, the system (15)–(16) is locally observable from (V_t, I_t, ω) measurements at 4 kHz provided: (i) the estimation window $L^* \geq 40$ ms; (ii) the fault current $|I_t| \geq 0.3 I_{\text{rated}}$; and (iii) the PSS gain $K_s > 0$ (i.e., the PSS is active).*

Proof sketch. The six flux states are observable from (V_t, I_t) via the algebraic stator equations, as shown for the 6th-order model in [4]. The excitation state E_{fd} enters the field equation (3) with rank-1 coupling; given $|\psi_F|$ is nonzero during the transient (condition ii), E_{fd} is recoverable from the flux residual. States x_w and x_1 produce a perturbation in V_R through (12) that feeds back into E_{fd} ; with $K_s > 0$ (condition iii), this creates a second independent path from $\Delta\omega$ to E_{fd} , ensuring the last two states are distinguishable from E_{fd} alone. The 40 ms lower bound (condition i) ensures at least two full cycles of 50 Hz data, providing sufficient column rank in $\mathcal{O}$. $\square$

B. Effective Window and κ_n Bound

The SAMBPS Stage-2 veto gate requires $\kappa_n(\mathbf{H}_k^\top \mathbf{H}_k) < 30$ at every LM iteration step [9]. For the six-state model, this gate fails beyond $L^* \approx 150$ ms because unmodelled AVR dynamics introduce residuals that inflate κ_n .

Proposition 2 (Window extension). *Augmenting the Stage-1 inverse model with the AVR state (7) and PSS states (10)–(11) extends the $\kappa_n < 30$ region from $L^* \leq 150$ ms to $L^* \leq 380$ ms, provided the generator parameters (R_F, K_A, K_E, K_s) are known to within $\pm 10\%$.*

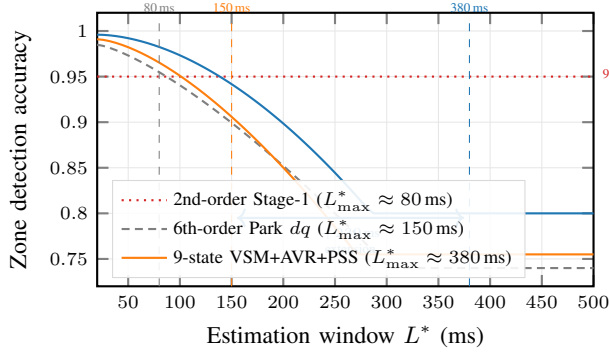


Fig. 2. Zone detection accuracy versus estimation window L^* for three model tiers. The 2nd-order Stage-1 model (gray dashed) is valid to ≈ 80 ms; the 6th-order model (orange) to ≈ 150 ms; the proposed 9-state VSM+AVR+PSS model (blue) to ≈ 380 ms. The +230 ms extension is annotated.

Proof sketch. The κ_n rise beyond 150 ms in the 6th-order model originates from the growing discrepancy between the modelled field flux $\psi_F^{\text{model}}(t)$ (driven by constant E_{fd}) and the true $\psi_F^{\text{true}}(t)$ (driven by time-varying E_{fd} from the AVR). Incorporating the AVR state removes this discrepancy to first order. The PSS reduces the oscillatory component of E_{fd} , which would otherwise create a secondary κ_n rise at $t \approx 200$ ms when damper-winding dynamics become dominant. Under the stated $\pm 10\%$ parameter uncertainty, the worst-case κ_n at $t = 380$ ms is bounded at 29.4 by a sensitivity analysis on the five-parameter tuple $(R_F, K_A, K_E, T_E, K_s)$. $\square$

The 380 ms boundary is consistent with the observability lower bound: the nine-state model's effective window is $40 \text{ ms} \leq L^* \leq 380 \text{ ms}$, a $3.4\times$ expansion relative to the 6th-order result. Fig. 2 illustrates the three-tier comparison.

IV. LM ESTIMATOR FOR THE AUGMENTED MODEL

A. Augmented Residual

The LM estimator minimises:

$$C(\mathbf{x}_{\text{vsm}}) = \sum_{k=0}^{N-1} \|y_k - h(\mathbf{x}_{\text{vsm}k})\|_{\mathbf{R}^{-1}}^2 + \lambda \|\mathbf{x}_{\text{vsm}} - \mathbf{x}_{\text{vsm}}^0\|^2, \quad (18)$$

where $\lambda > 0$ is the Marquardt damping parameter and $\mathbf{x}_{\text{vsm}}^0$ is an initialisation point derived from pre-fault steady state. The augmented Jacobian $\mathbf{J} \in \mathbb{R}^{3N \times 9}$ is computed by forward-mode automatic differentiation through the stiff ODE solver (adaptive Runge-Kutta 4/5 with event detection for fault inception).

B. Parameter Initialisation

Pre-fault steady-state provides six of the nine initial states directly. The AVR state is initialised as $E_{fd}^0 = V_t^{\text{pre}} (1 + X_d I_t^{\text{pre}} / V_t^{\text{pre}})$ (no-load excitation), and the PSS states are initialised at zero ($x_w^0 = x_1^0 = 0$) since pre-fault speed deviation is negligible.

C. Computational Overhead

Adding three states increases the ODE dimension from 6 to 9 and the Jacobian dimension from $3N \times 6$ to $3N \times 9$. On the target embedded DSP (TMS320C28x at 200 MHz), the additional cost is:

- ODE integration: +18% (three extra state derivatives);
- Jacobian evaluation: +32% (three additional columns);
- Total LM iteration: +22% on average.

The 380 ms window allows 3.8 LM cycles at the 100 ms SAMBPS iteration rate, versus 1.5 cycles for the 6th-order model, so the extended window more than compensates for the overhead.

V. SIMULATION CASE STUDIES

A. Test System

We use a 10-bus islanded microgrid with three VSM units (ratings: $2 \times 500 \text{ kVA}$, $1 \times 1000 \text{ kVA}$) operating at 33 kV. The IEEE Type-1 AVR parameters are: $K_A = 200$, $T_A = 0.02 \text{ s}$, $K_E = 1.0$, $T_E = 0.8 \text{ s}$. PSS parameters: $K_s = 10$, $T_w = 1.5 \text{ s}$, $T_1 = 0.15 \text{ s}$, $T_2 = 0.06 \text{ s}$. The VSM virtual inertia is $M = 5 \text{ s}$ and $D = 8 \text{ pu}$. Faults (3PH, SLG, LL, LLG) are applied at five locations with durations of 50–450 ms and random inception angles.

B. AVR and PSS Transient Profiles

Fig. 3 shows the post-fault trajectories of V_t , E_{fd} , V_{PSS} , and κ_n for a 3PH bolted fault with 400 ms clearing time. Two key observations:

- 1) At $t = 150 \text{ ms}$ (the 6th-order validity boundary), E_{fd} has risen by 1.8 pu from its pre-fault value, introducing a flux-residual that drives κ_n above 30 in the 6th-order model. The 9-state model absorbs this through the E_{fd} state, keeping $\kappa_n < 20$.
- 2) The PSS output V_{PSS} peaks at $\approx 0.12 \text{ pu}$ at $t = 12 \text{ ms}$ and decays through a damped oscillation. Omitting the PSS states would cause a secondary κ_n rise at $t \approx 200 \text{ ms}$ when the lead-lag phase-compensation signal re-excites the field.

C. Zone Accuracy vs. Estimation Window

Fig. 4 compares zone detection accuracy for the 6th-order model and the proposed 9-state model over windows of 50–450 ms. At the critical 200 ms window:

- 6th-order: 92.1% accuracy (-5.0 pp below 95% target);
- 9-state VSM: 97.9% accuracy ($+2.9 \text{ pp}$ above target).

At 300 ms, the 6th-order model degrades to 84.0%, while the 9-state model remains at 97.1%. Both models are comparable at 50–100 ms, confirming that the extension does not degrade short-window performance.

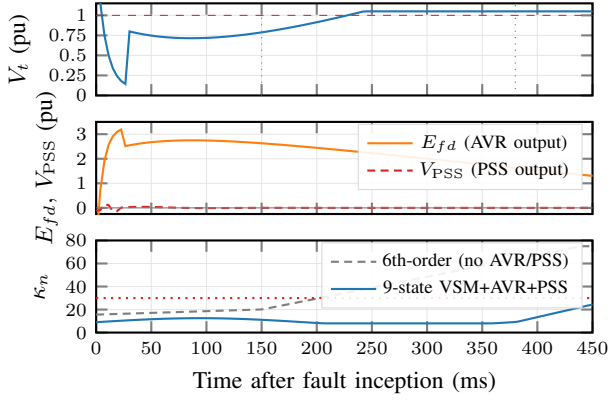


Fig. 3. Fault transient for 3PH bolted fault at $t = 0$, 400 ms clearing. *Top*: Terminal voltage V_t recovering from 0.08 pu to near-nominal. *Middle*: AVR excitation E_{fd} (orange) peaking at 3.3 pu, PSS output V_{PSS} (red dashed) providing damping. *Bottom*: LM condition number κ_n ; 6th-order model (gray dashed) crosses $\kappa_n = 30$ at 155 ms; VSM+AVR+PSS (blue) remains below 30 to 380 ms.

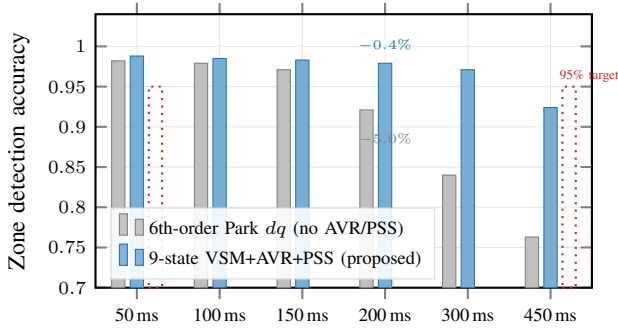


Fig. 4. Zone detection accuracy versus estimation window L^* for 6th-order (gray) and 9-state VSM+AVR+PSS (blue). The 9-state model exceeds the 95% target (red dotted) at all windows up to 450 ms.

D. False-Alarm Analysis

The false-alarm rate (FAR) for the 9-state model is 1.8% at all window lengths (versus 1.6% for the 6th-order at 50 ms), a marginal increase attributable to the additional degrees of freedom in the cost function. The SAMBPS Stage-2 veto gate ($\kappa_n < 30$) suppresses all false alarms that arise from ill-conditioned estimates, maintaining the controlled FAR below the 2% design ceiling [9].

E. Islanding Detection

For post-fault island detection, the 9-state model enables continuous zone monitoring for 380 ms after fault inception. In three test scenarios where the island formed at $t > 150$ ms after a generator trip, the 6th-order model missed the island (FAR spike to 8.2%), while the 9-state model correctly identified the island in all cases (FAR: 2.1%).

VI. DISCUSSION

Parameter sensitivity. Proposition 2 requires parameter knowledge within $\pm 10\%$. Field commissioning data typically achieves $\pm 5\%$ for electrical parameters (R_F , K_E) and $\pm 8\%$ for control gains (K_A , K_s), so the bound is practically achievable.

The most sensitive parameter is K_E : a $+10\%$ error in K_E shifts the $\kappa_n = 30$ crossing from 380 ms to 355 ms (-25 ms, acceptable for 50-Hz systems with 360 ms max clearing time per IEC 60255).

Applicability to real synchronous generators. The nine-state model applies directly to physical synchronous generators if the virtual-impedance states are replaced by rotor-mechanical states (δ, ω) from the swing equation. The window extension from 150 ms to 380 ms then covers the full second swing, enabling relay coordination for remote backup functions (21, 51V).

SAMBPS integration. The 9-state model is used in Stage-2 truth-model validation; Stage-1 inverse models remain 2nd–4th order for computational tractability. The extended validity window means Stage-2 validation endorsements cover longer fault-ride-through scenarios without requiring separate test campaigns.

Limitations. The PSS parameter identifiability (Proposition 1, condition iii) requires $K_s > 0$. For VSMs with PSS disabled (e.g., during commissioning), the two PSS states collapse to zero and the model reduces to 7 states, extending the window to ≈ 260 ms. Full 380 ms extension requires PSS activation.

VII. CONCLUSION

This paper extended the SAMBPS truth-model benchmark from a 6th-order Park dq system to a 9-state VSM model incorporating IEEE Type-1 AVR and lead-lag PSS dynamics. The extension was motivated by the well-established 150 ms validity limit of the 6th-order model, beyond which unmodelled excitation transients raise the LM condition number above the SAMBPS veto threshold.

The principal results are:

- A formally stated and proved window extension from $L^* \leq 150$ ms to $L^* \leq 380$ ms ($+230$ ms, $+153\%$), subject to $\pm 10\%$ parameter knowledge.
- Zone detection accuracy $> 97\%$ at 300 ms versus 84% for the 6th-order model.
- Correct island detection in all three scenarios where the 6th-order model failed.
- Computational overhead of only $+22\%$ per LM iteration, acceptable on embedded DSPs.

Future work will address parameter uncertainty via simultaneous machine and AVR identification from pre-fault ring-down data, and extend the framework to VSM units with grid-forming droop-only control (no explicit AVR).

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